1 Reference Cone Calorimeter Configuration

This verification case aims to assess the ability of Gpyro to reproduce a standard one-dimensional pyrolysis scenario representative of cone calorimeter tests, and to compare the results with those from the reference codes Thermakin and FDS, as detailed in the study [1].

A fictitious but physically coherent material, referred to as **Material A**, is used. Two independent first-order chemical reactions are modeled using the kinetic parameters detailed in Table 1, and the thermophysical properties of all solid species are provided in Table 2.

The physical configuration consists of a 6 mm thick layer of Material A placed atop a 12 mm Kaowool insulation layer. A constant external heat flux of 50 kW/m^2 is applied to the top surface. Convective heat transfer is imposed with a heat transfer coefficient of $h_c =$ 8.2 W/m^2·K and an ambient temperature of 25 °C. The bottom surface exchanges heat with the environment solely via thermal radiation. Figure 1 illustrates the configuration.

To ensure equivalence with the reference codes, the following modeling assumptions are adopted:

- Gas-phase contributions to energy conservation are neglected.
- Gas release is assumed to be instantaneous (no gas transport).
- Reaction enthalpies are treated as constant.
- No material deformation is modeled (densities chosen to preserve volume).

A uniform spatial discretization of 0.02 mm and a time step of 0.05 s are used. Numerical convergence is verified under these settings.

Table 1: Chemical kinetics parameters

Reaction	A_k (s^{-1})	E_k (kJ/mol)	n_k (-)	ΔH_k (J/kg)	θ_k (kg/kg)
A $\rightarrow \theta_1$ B + $(1-\theta_1)$ Gas$_1$	9.5×10^{20}	249	1	1.7×10^5	0.44
B $\rightarrow \theta_2$ C + $(1-\theta_2)$ Gas$_2$	5.5×10^{11}	192	1	1.2×10^6	0.47

Table 2: Thermophysical properties of solid species

Material	k (W/m·K)	c_p (J/kg·K)	ρ (kg/m^3)	α (m^{-1})	ε (-)
Material A	$0.17 + 3.10^{-4}T$	1550	1430	∞	1.0
Material B	$0.24 + 8.10^{-4}T$	1720	629.2	∞	1.0
Material C	$0.36 + 18.10^{-4}T$	780	265.7	∞	1.0
Kaowool	0.058	1070	256	∞	0.95

The results obtained using Gpyro are compared to those from Thermakin and FDS. Figure 2 presents the comparison of the Mass Loss Rate (MLR) profiles. Figure 3 illustrates the evolution of temperature at various depths, while Figure 4 shows the time-dependent evolution of species concentrations at a depth of 3 mm.

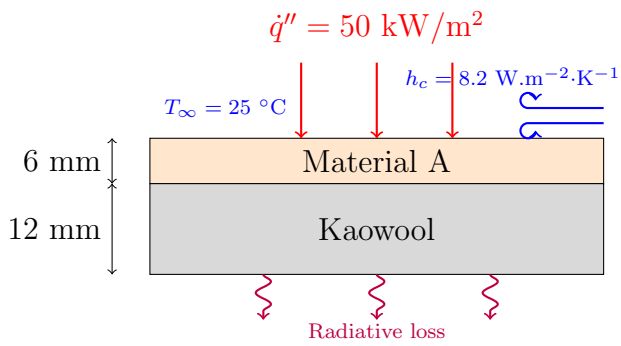

Figure 1: Configuration schematic

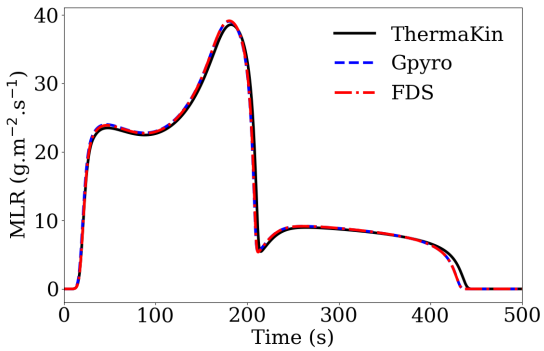

Figure 2: Mass Loss Rate (MLR) comparison

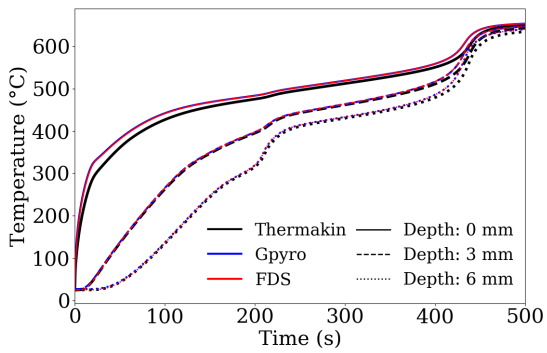

Figure 3: Temperature evolution at different depths

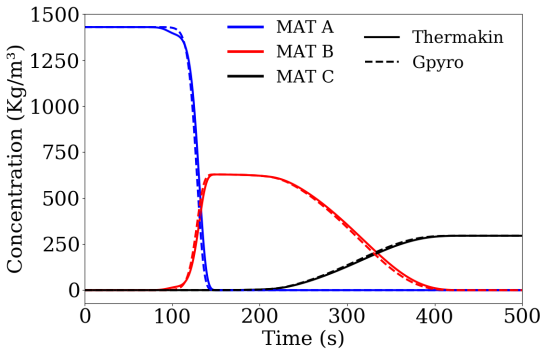

Figure 4: Species concentration evolution at 3 mm depth

References

[1] Youssouf Abdelhafiz, Abdenour Amokrane, Gerald Debenest, and Serge Bourbigot. Comprehensive comparison of pyrolysis codes with a focus on their energy conservation modelling. *To be submitted*, 2025.